

Mohammed Salah Elnaggar

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Portfolio: mohamedsalah112254-cell.github.io/Mechanical-BIM-Portfolio

EDUCATION

- **Professional Master's in sustainable water resources management**, Faculty of Engineering, Alexandria University, 2022-2024.
 - Awarded PG Scholarship managed by the American University in Cairo and funded by USAID.
 - Current GPA: 3.52 (B+).
- **Bachelor of Engineering**, Mechanical Engineering Department, Faculty of Engineering, Alexandria University, 2018-2022.
 - Grade: Good (70%)
- **Mechanical BIM Diploma**, KAITECH Academy, 02/2025- 08/2025
 - **Skills Gained:**
 - Detailed knowledge of BIM concepts and applications in mechanical projects
 - Hands-on experience with Revit and Dynamo
 - Enhanced teamwork abilities and coordination among specialists
 - BIM coordination, clash detection, information management, and sustainable design practices.
 - Mastery of 4D and 5D simulations, project lifecycle management
 - Integration of sustainable practices in designs and project management.

EXPERIENCE

[A&M International General Contracting](#), Cairo, Egypt (Head Office: EMIRATES). DEC 2025 - Present
Mechanical Technical Office Engineer.

- Study and review project models to verify alignment with project specification, BOQ, codes, and LOD 300.
- Manage model exchanges and documentation in **Autodesk Construction Cloud (ACC)** following CDE workflows
- Provide technical input during design reviews and resolve design issues promptly.
- Prepare accurate technical documents, drawings, and layouts.
- Develop and deliver MEP Revit models for HVAC and plumbing to LOD 400 in line with project BEP requirements.
- Ensure that designs meet sustainability, safety, and efficiency goals.
- Coordinate weekly multidisciplinary meetings; ensure models meet **ISO 19650** naming, classification, and information requirements.

Projects:

Rivage Tower, Reem Island, Abu Dhabi, EMIRATES.

- Project details: infrastructure facilities covering over 45,735 square meters.
levels: 2GR +2 PODIUM + 27 FLOORS + ROOF.
 - Shop drawing for MEP systems, especially HVAC in compliance with international standards and project specifications.
 - Model segmentation approach to support coordination and project requirement.
 - Prepare accurate technical documents, drawings, models, and layouts.
 - Provide technical input during design reviews and resolve design issues promptly.

Mechanical Design Engineer.

- Develop and design MEP systems, especially firefighting and plumbing, in compliance with international standards and project specifications.
- Prepare technical documents, drawings, models, and layouts.
- Ensure that designs meet sustainability, safety, and efficiency goals.
- Coordinate with other disciplines (architectural, structural, and electrical) to ensure seamless project integration.
- Utilize BIM software to maintain a cost-effective approach for projects.
- Provide technical input during design reviews and resolve design issues promptly.

Projects:

Overland Industrial Area, Doha, Qatar.

- Project details: infrastructure facilities covering over 237,000 square meters, including mosques, administrative buildings, and warehouses.
 - Develop and design MEP systems, especially firefighting and plumbing, in compliance with international standards and project specifications.
 - Ensure that designs meet sustainability, safety, and efficiency goals.
 - Prepare accurate technical documents, drawings, models, and layouts.
 - Provide technical input during design reviews and resolve design issues promptly.

Tarek Bin Ziad School, Qatar.

- Project details: Conceptual design of the firefighting system
 - Firefighting system concept based on a combined sprinkler and standpipe network, adapted to Qatar Civil Defense requirements for educational facilities.
 - Developed schematic layouts, zoning strategy, and technical documentation to support future detailed modeling.
 - Hydraulic calculations performed to validate pressure zones, flow rates, and pipe sizing across sprinkler and standpipe branches.
 - Ensure that designs meet sustainability, safety, and efficiency goals.

Bling Mall, New Administrative Capital, Egypt.

- Project details: Design and modeling of the firefighting and plumbing system.
 - Fire Fighting system modeling in compliance with international standards and project specifications.
 - Plumbing system modeling in compliance with international standards and project specifications.
 - Prepare accurate technical documents, drawings, models, and layouts.
 - Ensure that designs meet sustainability, safety, and efficiency goals.

Perfect Co., for Electromechanical Systems, Alexandria, Egypt.

MAY 2024- MAY 2025

Mechanical Technical Office Engineer.

- Design, review, and create shop drawings in accordance with design requirements and specifications.
- Utilize BIM software to maintain a cost-effective approach for projects by employing Revit and Navisworks.
- Coordinate with the consulting firms to submit the shop drawings for approval.
- Issue coordination drawings.
- Preparing technical requirements such as calculations and BOQ.
- Drafted detailed dimensional drawings and designs using AutoCAD.

Projects:

Dahshur Hangers, Cairo, Egypt.

- Project details: 10 hangers with double levels, each hanger with 10000 m2.
 - Fire Fighting & plumbing shopdrawing according to Egyptian code.
 - Prepare shop drawing submissions for approval.
 - Site management to ensure that works aligned with shop drawing and project specifications.

Custom Building Extension, Alexandria Port, Egypt.

- Project details: Design the HVAC system and firefighting system.
 - Fire Fighting modeling according to the Egyptian code.
 - Calculate the air-conditioning load with HAP software.
 - Calculate ventilation requirements according to ASHRAE 62.1.
 - Prepare shop drawing submissions for approval.

The Center for Applied Research on the Environment and Sustainability, AUC in Cairo

AUG -2022

Internship Trainee.

- I had an opportunity to delve into the water resources industry, focusing specifically on the water industry in Egypt, and the Water-Energy-Food (WEF) Nexus. It allowed me to gain insights into Egypt's water scarcity and status by demonstrating and evaluating key concepts:
 - The concept of sustainable development through the Development Goals (SDGs).
 - Water scarcity, virtual water, and water footprint.

Graduation Project

A solar-powered reverse osmosis desalination plant with a pressure exchanger.

JULY 2022

- Development of seawater and wells reverse osmosis desalination plant with a capacity of 5m³ per day using an innovative design of pressure exchanger that recovers up to 60% of power by decreasing the flow of the high-pressure pump and modeling the whole plant on the computer to detect the effect of any change that may occur to any of the input parameters of the plant. By using solar energy power saving may reach 80% of total power consumption.

Achievements:

- Awarded Best Project in the Fluid Division of the Mechanical Department in 2022.
- Funded by the Egyptian Academy of Scientific Research and Technology with a grant of 75,000 EGP.

Project supervisors:

- Dr. Ihab Adam, DR. Mohamed Sewdane, Eng. Mohamed Fouad.

Courses

- Navisworks Manage by Eng – Mohamed Gamal.
- Revit MEP for Mechanical Engineers by Autodesk.
- Revit MEP by Eng - Ahmed Shuhayb.
- Project Management from the University Center for Career Development provided by AUC.
- Employability Skills Track from the University Center for Career Development provided by AUC.

Skills

Skills:

- **Programming Languages:** Python (Basics)
- Revit MEP (V.Good), AutoCAD (Excellent), Navisworks (V.Good), SolidWorks (Intermediate),
- Toray DS2 (V.Good), Water Gems (Good), Hammer V8i (V.Good), MS Office (Excellent),
- HTRI (Excellent), HAP (V.Good), Duct Sizer (Excellent), MATLAB Simulink (Good).

Language Skills:

- English: Very good at writing and listening, and good at speaking.
- Arabic (Native).

Personal Information

Nationality: Egyptian.

Military service: completed.